

DELIMITER \$\$

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/* =====
PROCEDURE 1: CALCULATE CUMULATIVE GPA
===== */
CREATE PROCEDURE CalculateGPA(IN p_student_id VARCHAR(10))
BEGIN
    DECLARE total_points DECIMAL(10,2) DEFAULT 0;
    DECLARE total_credits INT DEFAULT 0;
    DECLARE calculated_gpa DECIMAL(5,2);

    -- Get total grade points and credits
    SELECT
        IFNULL(SUM(g.grade_point * c.credits), 0),
        IFNULL(SUM(c.credits), 0)
    INTO total_points, total_credits
    FROM enrollment e
    JOIN grade g ON e.grade_letter = g.grade_letter
    JOIN course c ON e.course_id = c.course_id
    WHERE e.student_id = p_student_id;

    -- Calculate GPA
    IF total_credits > 0 THEN
        SET calculated_gpa = total_points / total_credits;
    ELSE
        SET calculated_gpa = 0;
    END IF;

    -- Insert new GPA record (log style)
    INSERT INTO gpa_record (
        student_id,
        semester,
        academic_year,
        semester_gpa,
        cumulative_gpa,
        total_credits_earned
    )
    VALUES (
        p_student_id,
        'LATEST',
        'CURRENT',
        calculated_gpa,
        calculated_gpa,
        total_credits
    )
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);
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END $$
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/* =====  
PROCEDURE 2: CALCULATE TARGET GPA  
===== */  
CREATE PROCEDURE CalculateTargetGPA(  
    IN p_student_id VARCHAR(10),  
    IN p_target_gpa DECIMAL(3,2)  
)  
BEGIN  
    DECLARE completed_credits INT DEFAULT 0;  
    DECLARE current_points DECIMAL(10,2) DEFAULT 0;  
    DECLARE total_program_credits INT DEFAULT 120; -- fallback  
    DECLARE remaining_credits INT;  
    DECLARE required_points DECIMAL(10,2);  
    DECLARE required_gpa DECIMAL(5,2);  
  
    -- Try to get program total credits (if linked later)  
    SELECT p.total_credits  
    INTO total_program_credits  
    FROM programme p  
    LIMIT 1;  
  
    -- Get completed credits and points  
    SELECT  
        IFNULL(SUM(c.credits), 0),  
        IFNULL(SUM(g.grade_point * c.credits), 0)  
    INTO completed_credits, current_points  
    FROM enrollment e  
    JOIN course c ON e.course_id = c.course_id  
    JOIN grade g ON e.grade_letter = g.grade_letter  
    WHERE e.student_id = p_student_id;  
  
    -- Remaining credits  
    SET remaining_credits = total_program_credits - completed_credits;  
  
    -- Required total points  
    SET required_points = p_target_gpa * total_program_credits;  
  
    -- Required GPA calculation  
    IF remaining_credits > 0 THEN
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        SET required_gpa = (required_points - current_points) / remaining_credits;
ELSE
    SET required_gpa = 0;
END IF;

-- Output result
SELECT
    p_student_id AS student_id,
    p_target_gpa AS target_gpa,
    required_gpa AS required_gpa_needed,
    remaining_credits;

END $$

DELIMITER ;

CALL CalculateGPA('STU001');
CALL CalculateTargetGPA('STU001', 3.50);
```